ReqRes

REQRES API TESTING PROJECT

TEST PLAN DOCUMENT

API Testing using Postman

Created By

Sanika Kale

Document Version: 1.0

September 2026

Document Control

Document Item	Details
Project Name	ReqRes API Testing Project
Document Name	API Test Plan
Version	1.0
Prepared By	Sanika Kale
Testing Tool	Postman
Scripting Language	JavaScript
API Type	REST API
Test Environment	ReqRes Public API
Status	Portfolio Version

Table of Contents

1. Introduction
2. Project Overview
3. Objectives
4. Scope of Testing
5. Out of Scope
6. API Modules and Endpoints
7. Test Strategy
8. Types of Testing
9. Test Environment
10. Tools and Technologies
11. Test Data
12. Entry Criteria
13. Exit Criteria
14. Test Execution Process
15. Defect Management
16. Risk and Mitigation
17. Deliverables
18. Assumptions and Dependencies

Introduction

The ReqRes API Testing Project is designed to validate the functionality, reliability, and behavior of REST APIs using Postman. The project focuses on testing different HTTP methods, including GET, POST, PUT, PATCH, and DELETE, to verify that API requests and responses work as expected. The testing process includes validating request parameters, headers, request bodies, response bodies, HTTP status codes, and JSON response structures. Both positive and negative test scenarios are included to verify successful operations as well as proper handling of invalid requests and data. JavaScript assertions in Postman are used to automate response validations and reduce manual verification. The project also validates important response fields, data types, headers, and response times. Detailed test cases are prepared to provide complete coverage of the identified API functionality. Test execution results are recorded with actual results, pass/fail status, priority, and comments. Any unexpected behavior identified during testing is documented in a defect report for further analysis. Overall, this Test Plan defines the testing approach, scope, objectives, resources, deliverables, and criteria required to systematically test the ReqRes APIs.

2. Project Overview

The ReqRes API Testing Project focuses on testing REST APIs using Postman. The project covers GET, POST, PUT, PATCH, and DELETE requests, along with positive and negative test scenarios. API responses are validated using status codes, response bodies, headers, JSON structure, and JavaScript assertions. Test cases and execution results are documented to ensure complete API functionality and reliability.

3. Objectives

The objectives are to verify API functionality; validate HTTP status codes; validate JSON response structure and fields; validate request and response headers; perform positive and negative testing; test GET, POST, PUT, PATCH and DELETE methods; implement Postman JavaScript assertions; and maintain professional API test documentation.

4. Scope of Testing

- GET APIs: Validate user listing, single-user details, pagination, and invalid user requests.
- POST APIs: Validate user creation with valid, missing, empty, and invalid request data.
- PUT APIs: Validate complete user updates and invalid update requests.
- PATCH APIs: Validate partial user updates and invalid requests.
- DELETE APIs: Validate user deletion and invalid delete requests.
- Response Validation: Verify status codes, response body, JSON structure, headers, and required fields.
- Negative Testing: Validate API behavior for invalid endpoints, data, IDs, and request bodies.
- Automation: Use Postman JavaScript assertions for response validation.

5. Out of Scope

- UI/Frontend testing
- Database testing
- Performance and load testing
- Security penetration testing
- Mobile application testing
- Infrastructure/server testing
- Production environment testing
- End-to-end business workflow testing
- Third-party API integration testing
- Automated performance testing

6. API Modules and Endpoints

The project covers user-related REST operations. Representative endpoints include /api/users?page=2, /api/users/2, /api/users, and invalid-resource scenarios. Current service behavior should be verified at execution time because public practice APIs may change.

7. Test Strategy

Each test case defines the endpoint, HTTP method, URL, headers, request body, expected response, expected status code, and validation criteria. The tester sends the request, validates status and response structure, validates data fields, executes JavaScript assertions, records the actual result, and marks Pass/Fail.

8. Types of Testing

Functional Testing verifies endpoint behavior. Positive Testing uses valid requests. Negative Testing uses invalid IDs, endpoints, fields, bodies, and JSON. Status Code Validation checks expected HTTP codes. Response Validation checks JSON structure, fields and values. Header Validation checks important headers. Response Time Validation performs basic timing checks. Regression Testing re-runs affected cases after changes.

9. Test Environment

Testing is performed against the ReqRes public API over HTTPS using Postman. A working internet connection is required. Execution should be performed against the current service behavior and the execution date should be recorded.

10. Tools and Technologies

Postman for API execution and collections; JavaScript for Postman assertions; Microsoft Excel for test cases and execution results; Microsoft Word for documentation; GitHub for source control and portfolio presentation.

11. Test Data

Test data includes valid IDs such as 1 and 2, non-existent IDs such as 9999, valid names and job values, updated values, empty request bodies, missing fields, invalid JSON, and invalid endpoint paths. No confidential credentials or sensitive personal data should be used.

12. Entry Criteria

ReqRes must be accessible; Postman must be installed and working; test cases must be prepared; endpoints and request formats must be available; test data must be ready; and internet connectivity must be available.

13. Exit Criteria

All planned test cases should be executed; actual results and Pass/Fail values should be recorded; failures should be investigated; defects should be documented where applicable; execution evidence should be saved; and final documentation should be reviewed.

14. Test Execution Process

Prepare the request in Postman, configure headers and body, send the request, validate status code, validate headers and JSON structure, validate data fields, execute JavaScript assertions, record actual results, investigate failures, and re-test applicable cases.

15. Defect Management

Unexpected behavior should be documented when observed behavior does not match the defined expected behavior or current documented API behavior. A defect should include ID, test case ID, summary, endpoint, method, steps, request, expected result, actual result, status code, severity, priority, evidence, and status.

16. Risk and Mitigation

Public API behavior may change; verify current behavior before execution. Network issues may affect tests; confirm connectivity and re-run. Test data may behave differently; use representative alternatives. Incorrect assertions may cause false failures; compare scripts with actual response structures.

17. Deliverables

Deliverables include API Test Plan, API Test Case Excel, Postman Collection JSON, Postman JavaScript scripts, Test Execution Report, Defect Report when applicable, execution screenshots, and GitHub project documentation.

18. Assumptions and Dependencies

Assumptions include ReqRes availability, accessible endpoints, working Postman, and permission to execute requests against the public practice API. Dependencies include ReqRes availability, internet connectivity, Postman, Excel/Word, and GitHub.